RAMCO INSTITUTE OF TECHNOLOGY

RAJAPALAYAM-626117

Department of Artificial Intelligence and Data Science

Academic Year: 2025- 2026 (Odd Semester)

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science.

Course Code & Title: AD3301- Data Exploration and Visualization. Section: A

Name of the Faculty member: Dr.M.Kaliappan, Professor/AD

Theme of discussion: Exploratory Data Analysis phases.

Date and Time: 19.08.2025 & 1.55 pm to 2.45pm

Peer Instruction: An Innovative Peer Learning Practice

Peer Instruction (PI) is a learner-centered methodology that enhances conceptual understanding through structured peer-to-peer discussions. Unlike traditional lecture-based delivery, this approach emphasizes active participation, reflection, and dialogue, enabling students to construct knowledge collaboratively.

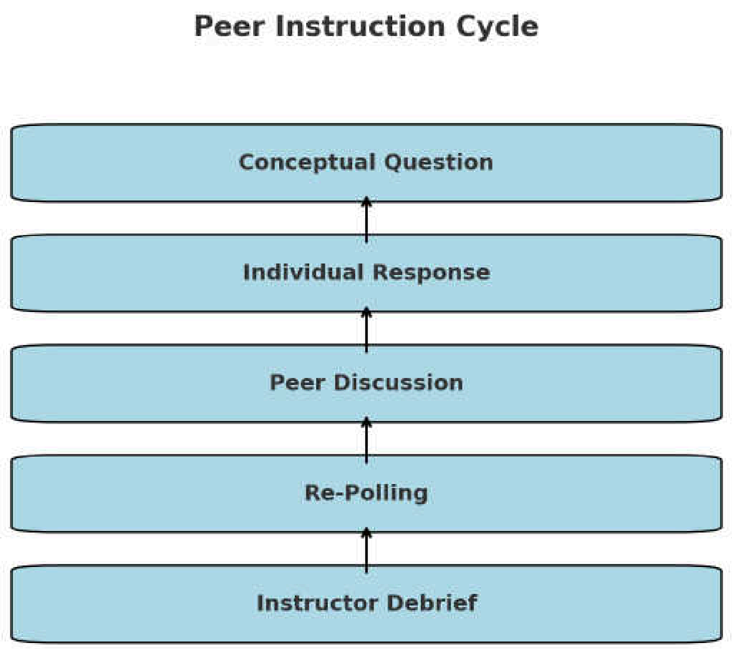
Objective

- To implement Peer Instruction as an innovative peer learning strategy.
- To enhance students' conceptual understanding of the phases of Exploratory Data Analysis (EDA)**.**
- To promote critical thinking, collaboration, and communication skills among learners.

Implementation Framework for the Session

1. **Conceptual Question:**
   The instructor introduces a thought-provoking question related to the phases of Exploratory Data Analysis (EDA), such as identifying the correct sequence of EDA tasks or interpreting a dataset summary.
2. **Individual Response:**
   Students independently attempt the question, recording their answers through polling tools or paper slips.

3. **Peer Discussion:**
   Learners engage in peer conversations, justifying their reasoning, debating interpretations, and clarifying misconceptions about data cleaning, visualization, and pattern detection in EDA.
4. **Re-Polling:**
   Students respond again after the peer discussion. Improvements in responses typically reflect enhanced conceptual clarity.
5. **Instructor Debrief:**
   The teacher facilitates a short discussion to consolidate learning, explain nuances in EDA phases, and connect the activity to real-world data analysis practices.



Observations

- Active participation was evident as all learners engaged in peer conversations.
- Peer discussions encouraged articulation of reasoning and improved conceptual clarity.
- Students reported greater confidence in applying EDA phases after collaborative learning.
- Real-time feedback enabled the instructor to identify common misconceptions.

Innovative Features

- Promotes critical thinking and collaborative problem-solving in data science contexts.
- Transforms the classroom into a discussion-driven, student-centered environment.
- Provides real-time feedback for both learners and instructors.
- Can be integrated with digital learning platforms for interactive analytics training.

Expected Outcomes

- Improved understanding of **EDA phases and their applications.**
- Enhanced ability to **interpret and communicate data insights.**
- Development of **teamwork and communication skills** essential for data-driven problem-solving.
- Fosters a **collaborative learning culture** that prepares students for advanced analytics and research tasks

19/8
Faculty Incharge

HOD/AD 19/8/25